H2O: Why Water Is Special

01 What does H2O mean?



Two hydrogen atoms and one oxygen atom join to make one water molecule.

Look closer. Find the small three-part model beside the glass: two matching pieces and one larger piece.

A little explanation. Water is made of tiny pieces called molecules. Each water molecule is H2O: two hydrogen atoms joined to one oxygen atom. Atoms are even smaller pieces that can join to make molecules.

Try this. Put one ice cube on a plate and start a timer. Use only cool water, keep a towel nearby, and wipe up every spill. Watch until the ice melts.

02 Why do water drops bead up?



The attraction between water molecules helps drops gather into rounded beads.

Look closer. Look at the rounded drops on wax paper and the larger drop where two small ones have joined.

A little explanation. Water molecules attract one another. At the surface, their pull creates surface tension, which helps a drop gather into a rounded bead instead of spreading flat.

Try this. Use a dropper or the end of a straw to place a few water drops on wax paper. Nudge two drops together and watch them join.

03 How can a paperclip rest on water?



Surface tension lets gently placed paperclips rest on the top of water.

Look closer. See how the paperclip rests across the surface while the water bends slightly beneath it.

A little explanation. Surface tension makes the top of the water resist being pulled apart. It can act a little like a skin, but it is not a real solid skin. A paperclip can rest there if it is placed gently.

Try this. Lay a small piece of tissue on water, place a clean paperclip on the tissue, and wait for the tissue to sink while the paperclip stays.

04 Why does ice float?



Ice is lighter for its size than liquid water, so it floats.

Look closer. Compare the ice cube above the waterline with the liquid below it and the winter pond beyond the window.

A little explanation. When water freezes, its molecules settle into a roomier pattern. Ice is lighter for its size than liquid water, so it floats. That unusual clue helps ponds freeze from the top while liquid water remains below.

Try this. Drop an ice cube into a clear cup of water. Look from the side and notice how part of the cube stays above the surface.